

0 4 . 1

Complete **Table 3** to show whether the feature of DNA is associated with the DNA molecule found in each of these locations.

Tick (✓) the appropriate boxes.

[3 marks]

Table 3

Feature of DNA	Location of DNA molecule		
	Prokaryotic cell	Nucleus	Chloroplast
Is circular			
Contains four different types of nucleotide			
Is associated with histones			

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Question 4 continues on the next page

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ANSWER IN THE SPACES PROVIDED**

Turn over ►



Some strains of the bacterium, *Escherichia coli*, are known to be resistant to antibiotics.

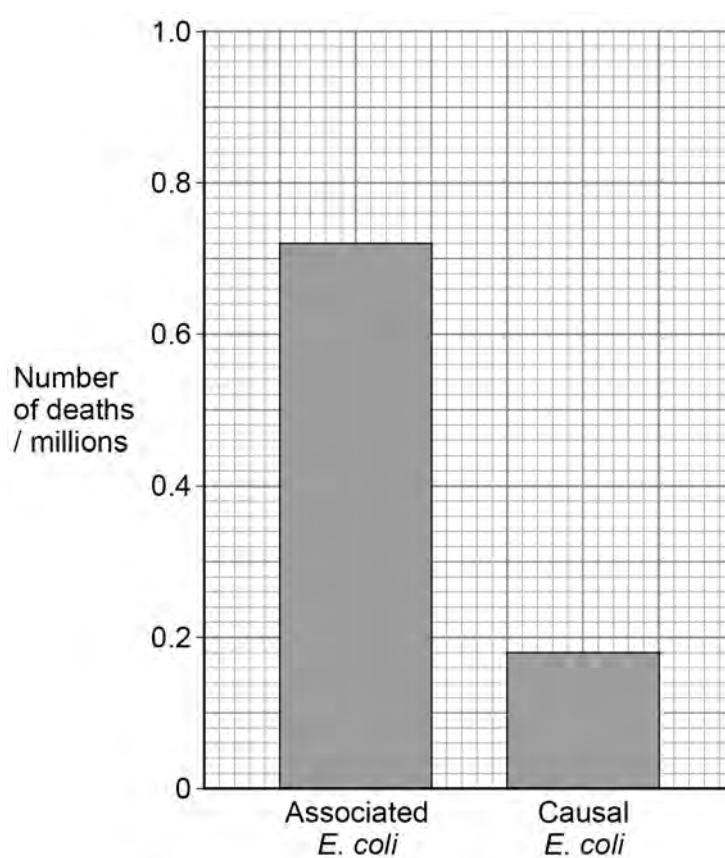
Scientists analysed data recording the deaths of people in hospitals across the world during 2019.

They used the data to find how many of the bodies of people who died in hospitals contained antibiotic-resistant *E. coli* bacteria and whether the *E. coli* were:

- **not** the cause of death (**associated** *E. coli*)
- the cause of death (**causal** *E. coli*).

Figure 5 shows their results.

Figure 5



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0 4 . 2

Using **Figure 5**, a student calculated that out of all the people who died when antibiotic-resistant *E. coli* were present, 25% of those deaths involved **causal** *E. coli*.

The student's calculation is **incorrect**.

Use **Figure 5** to calculate the correct percentage of deaths involving **causal** *E. coli* in people who died when infected with antibiotic-resistant *E. coli*.

Identify the mathematical step the student performed incorrectly in their calculation.

[3 marks]

Correct answer _____ %

Incorrect mathematical step _____

0 4 . 3

Which method is most likely to be successful in **decreasing** the frequency of antibiotic-resistant bacteria in populations of people?

Tick (✓) **one** box.

[1 mark]

Give people lower doses of antibiotics to treat disease.

☐

Test more people to determine if they are infected with antibiotic-resistant bacteria.

☐

Vaccinate more people to reach herd immunity against bacteria that cause diseases common in human populations.

☐

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